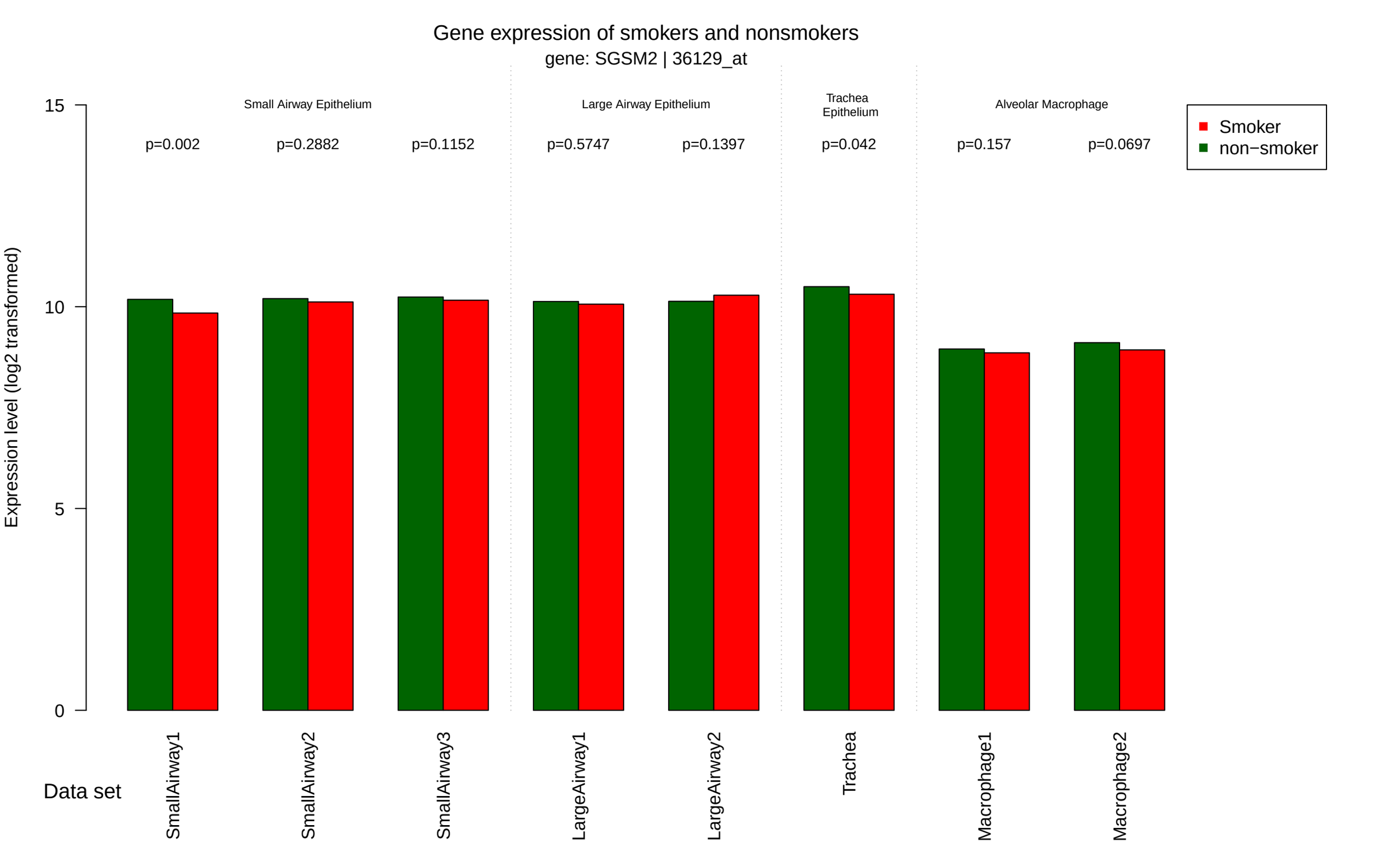
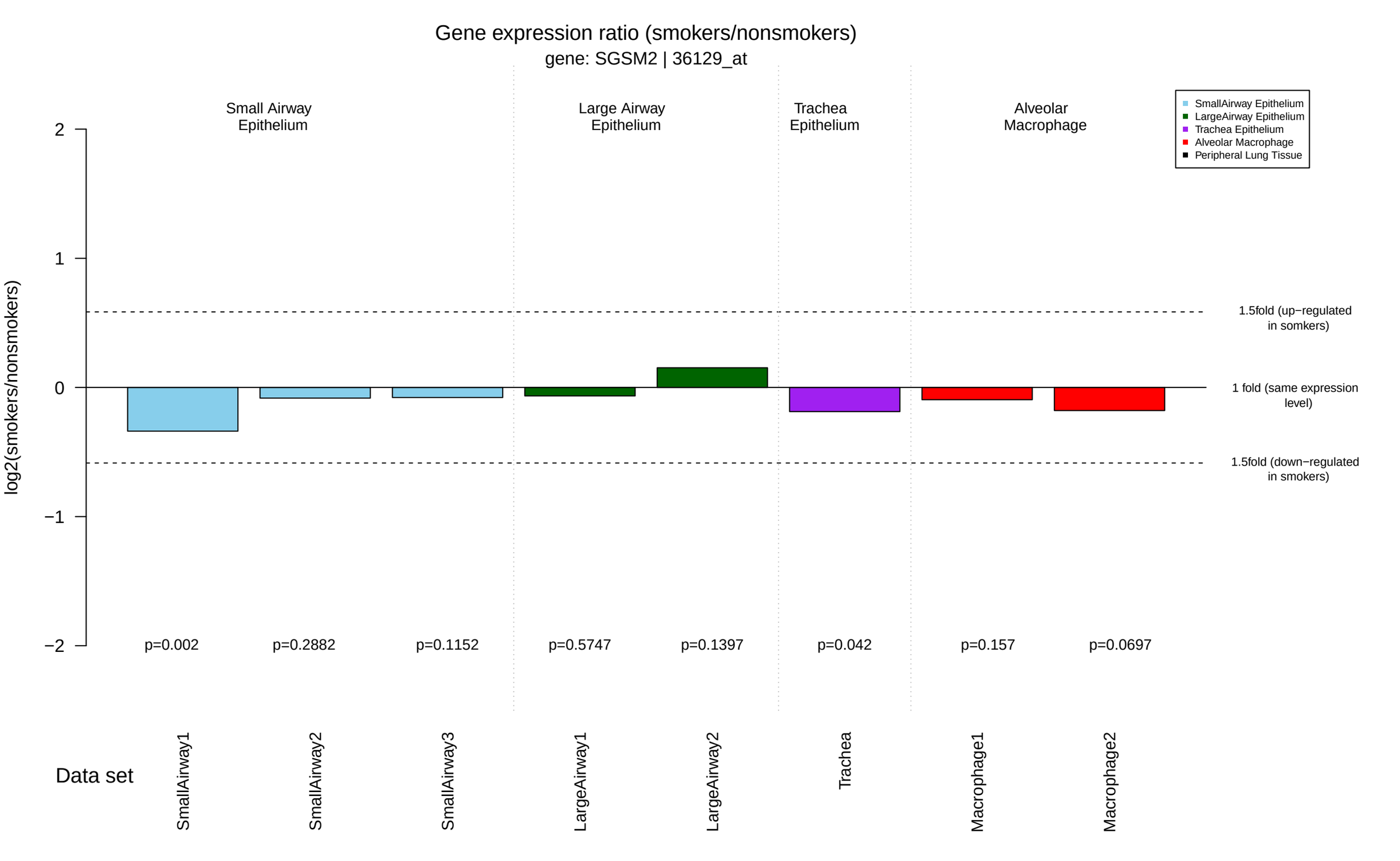
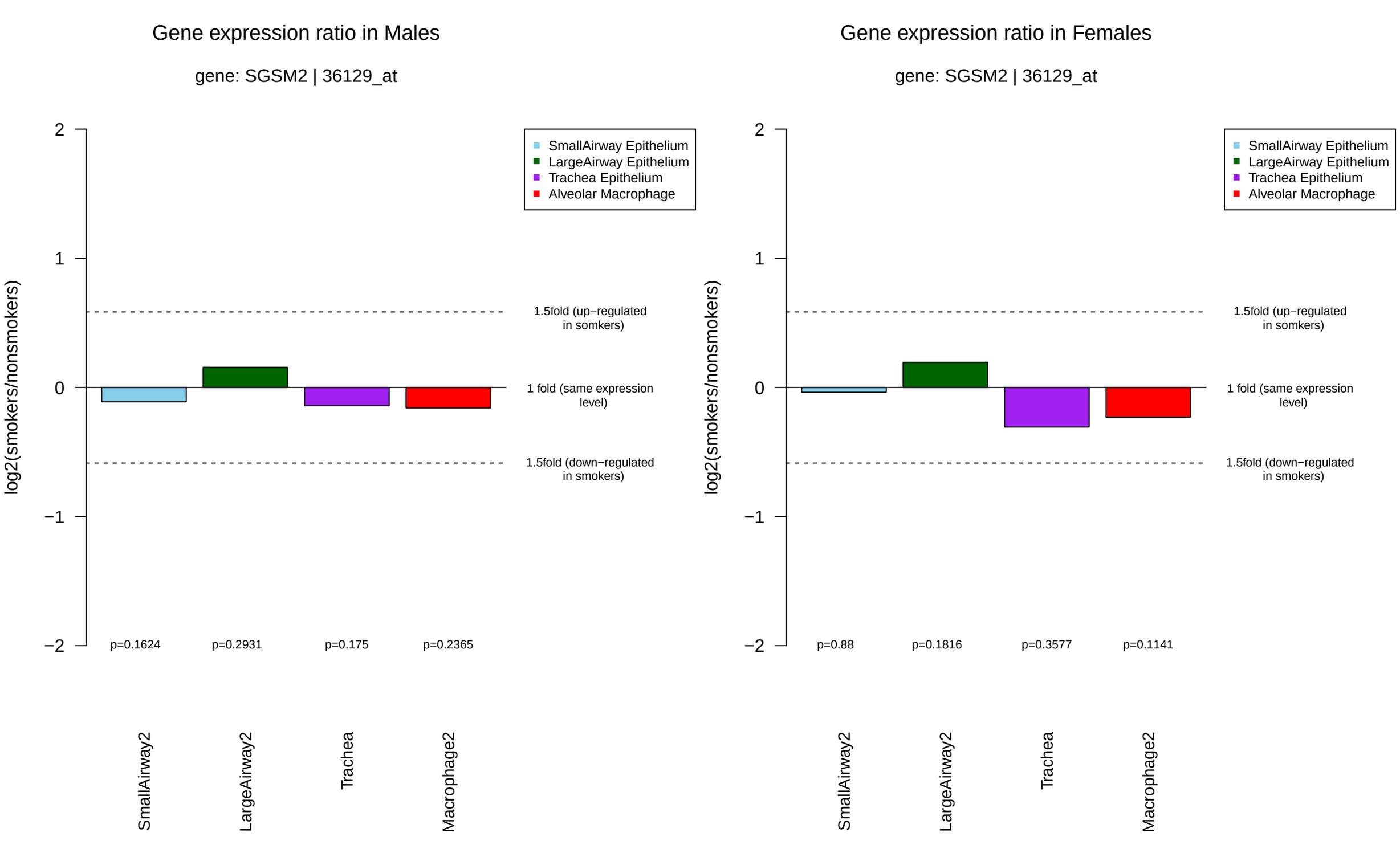
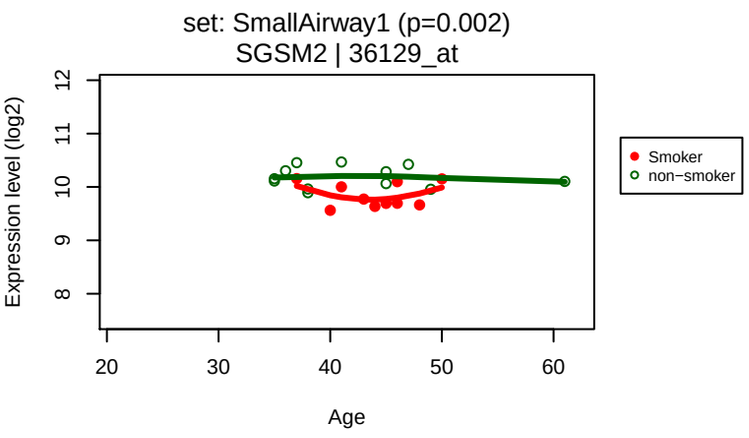
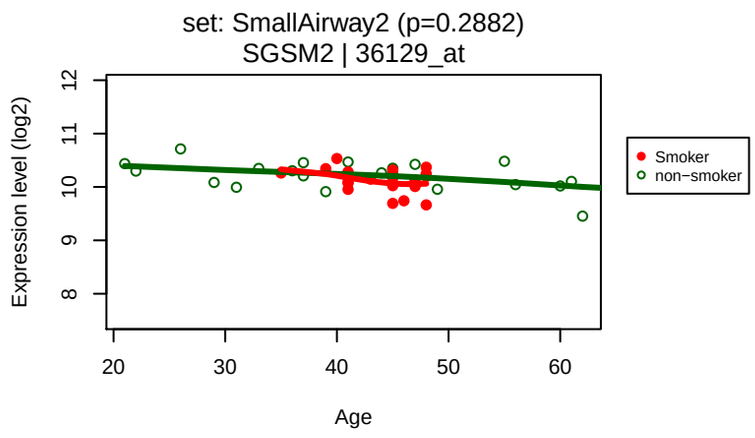
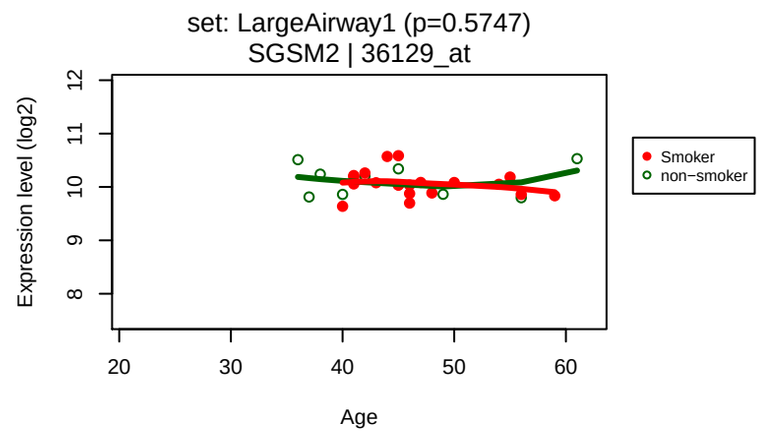

Gene expression of smokers and nonsmokers
gene: SGSM2 | 36129_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.002
p=0.2882
p=0.1152
p=0.5747
p=0.1397
p=0.042
p=0.157
p=0.0697
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

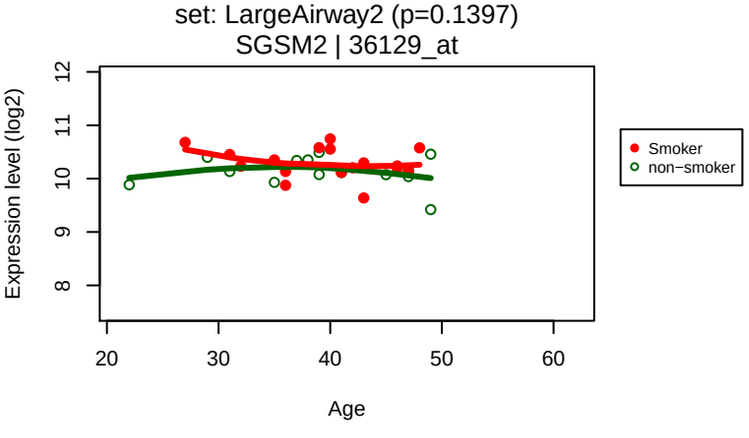

Gene expression ratio (smokers/nonsmokers)
gene: SGSM2 | 36129_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.002
p=0.2882
p=0.1152
p=0.5747
p=0.1397
p=0.042
p=0.157
p=0.0697
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

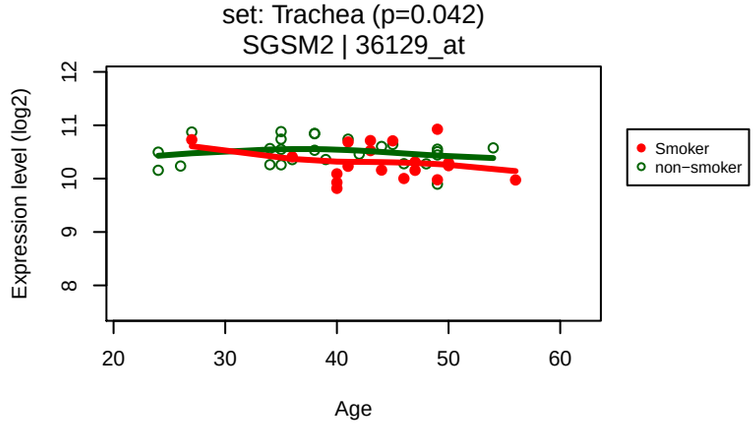

Gene expression ratio in Males
gene: SGSM2 | 36129_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1624
p=0.2931
p=0.175
p=0.2365
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SGSM2 | 36129_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.88
p=0.1816
p=0.3577
p=0.1141
SmallAirway2
LargeAirway2
Trachea
Macrophage2

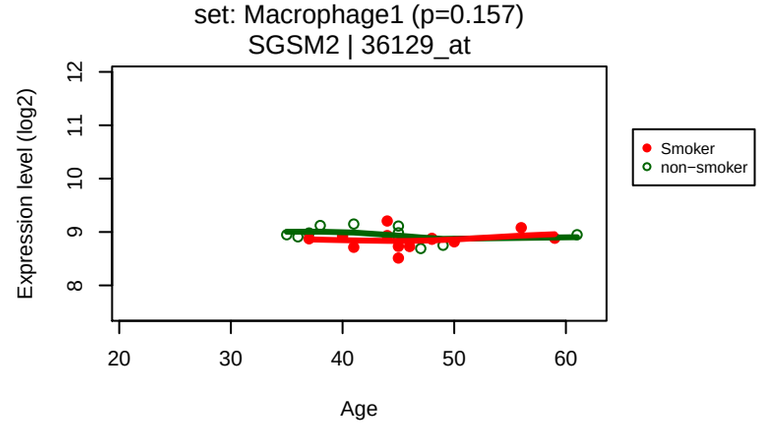

set: SmallAirway1 (p=0.002)
SGSM2 | 36129_at
Expression level (log2)
Age
Smoker
non−smoker

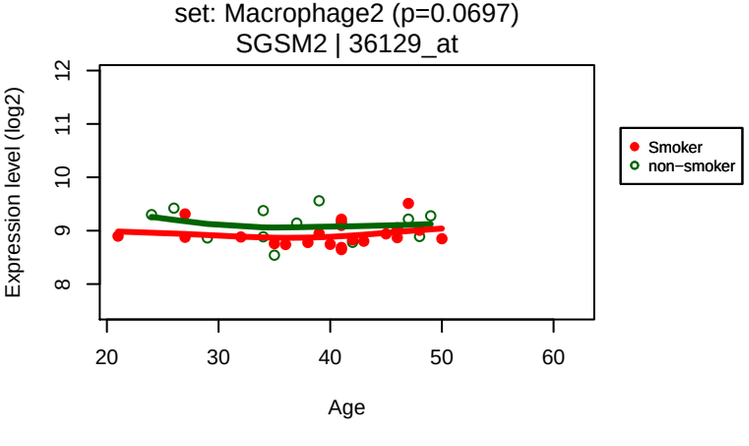

set: SmallAirway2 (p=0.2882)
SGSM2 | 36129_at
Expression level (log2)
Age
Smoker
non−smoker

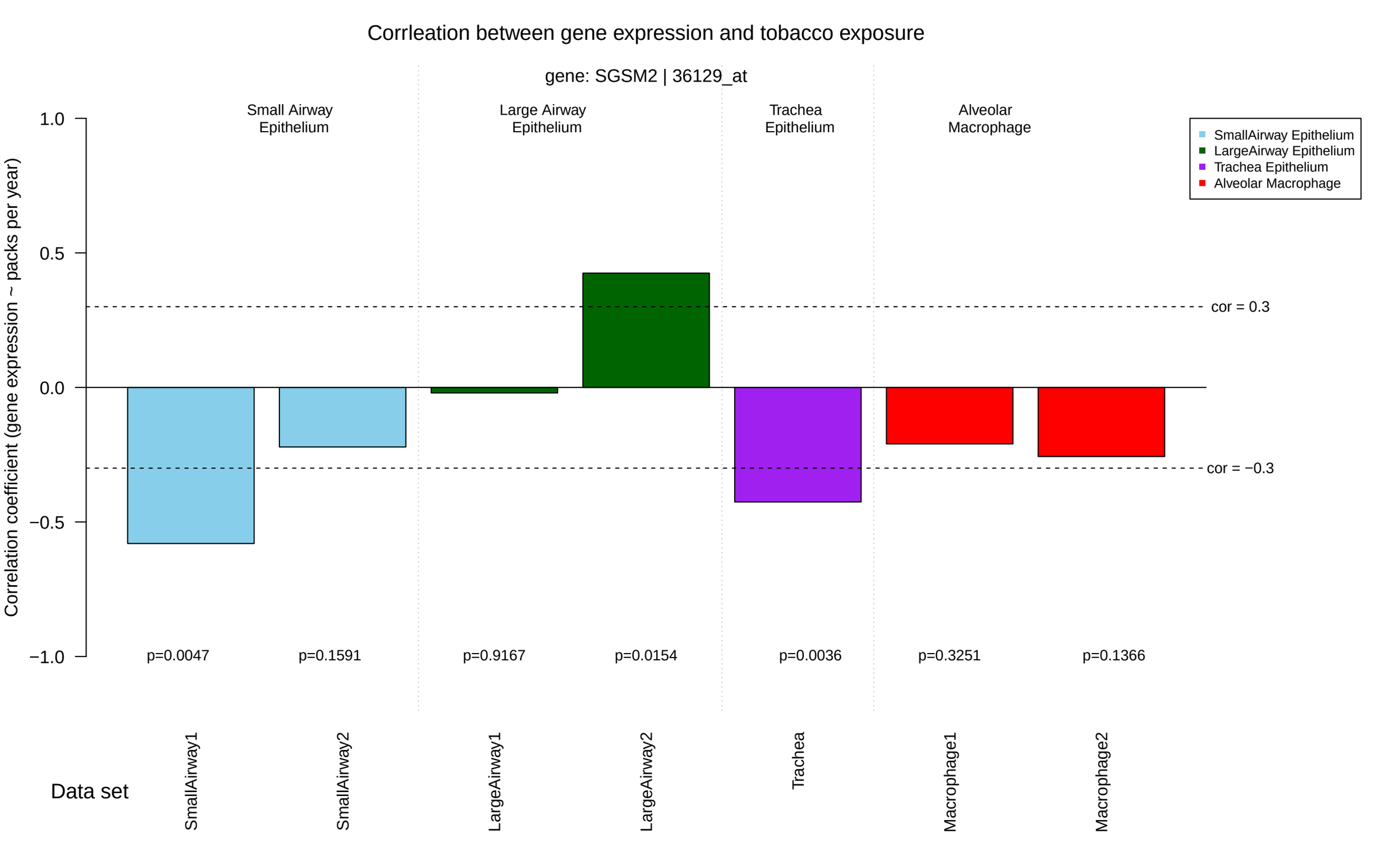

set: LargeAirway1 (p=0.5747)
SGSM2 | 36129_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1397)
SGSM2 | 36129_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.042)
SGSM2 | 36129_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.157)
SGSM2 | 36129_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0697)
SGSM2 | 36129_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SGSM2 | 36129_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0047
p=0.1591
p=0.9167
p=0.0154
p=0.0036
p=0.3251
p=0.1366
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2